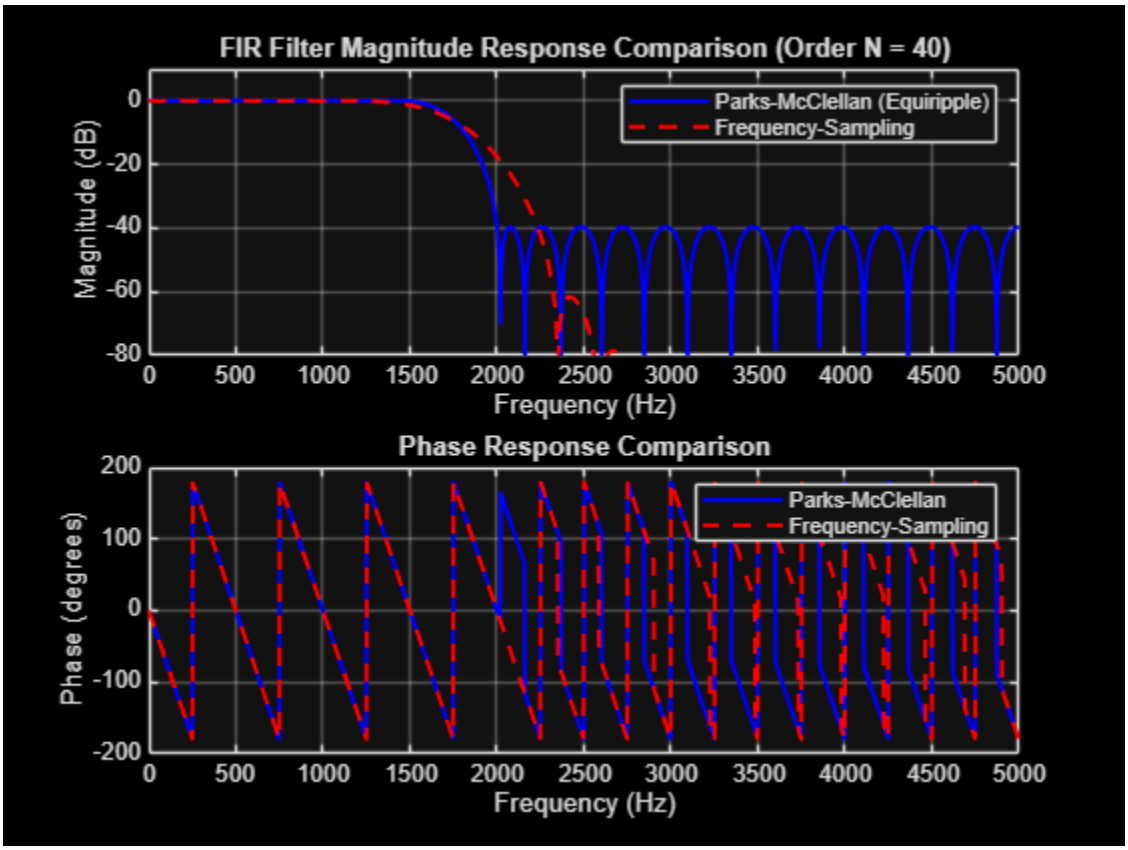
```
clear;
clc;
close all;

% EXPERIMENT 5: PARKS-MCCLELLAN vs FREQUENCY-SAMPLING

Fs = 10000;       % Sampling frequency (Hz)
fp = 1500;        % Passband edge (Hz)
fs = 2000;        % Stopband edge (Hz)
N = 40;           % Filter order

% ========================================
% 1. PARKS-MCCLELLAN (EQUIRIPPLE) DESIGN
% ========================================

f_pm = [0 fp fs Fs/2] / (Fs/2);
a_pm = [1 1 0 0];

b_pm = firpm(N, f_pm, a_pm);

% ========================================
% 2. FREQUENCY-SAMPLING DESIGN
% ========================================

f_fs = [0 fp fs Fs/2] / (Fs/2);
a_fs = [1 1 0 0];

b_fs = fir2(N, f_fs, a_fs);

% ========================================
% 3. DISPLAY NUMERICAL OUTPUTS
% ========================================

fprintf('=== EXPERIMENT 5: PARKS-MCCLELLAN vs FREQUENCY-SAMPLING ===\n\n');

fprintf('Filter Order (N): %d\n', N);
fprintf('Passband Edge: %d Hz\n', fp);
fprintf('Stopband Edge: %d Hz\n\n', fs);

fprintf('--- PARKS-MCCLELLAN COEFFICIENTS (First 10) ---\n');
disp(b_pm(1:10));

fprintf('--- FREQUENCY-SAMPLING COEFFICIENTS (First 10) ---\n');
disp(b_fs(1:10));

% ========================================
% 4. FREQUENCY RESPONSE
% ========================================

[H_pm, f] = freqz(b_pm, 1, 2048, Fs);
[H_fs, ~] = freqz(b_fs, 1, 2048, Fs);
```

```
% Avoid log10(0)
mag_pm = 20*log10(max(abs(H_pm), 1e-8));
mag_fs = 20*log10(max(abs(H_fs), 1e-8));

% ========================================
% 5. PLOTTING
% ========================================

figure('Name', 'Parks-McClellan vs Frequency-Sampling', ...
    'NumberTitle', 'off');

% Magnitude response
subplot(2,1,1);

plot(f, mag_pm, 'b', 'LineWidth', 1.5);
hold on;
plot(f, mag_fs, 'r--', 'LineWidth', 1.5);

grid on;
xlim([0 Fs/2]);
ylim([-80 10]);

legend('Parks-McClellan (Equiripple)', ...
    'Frequency-Sampling', 'Location', 'northeast');

title('FIR Filter Magnitude Response Comparison (Order N = 40)');
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');

% Phase response
subplot(2,1,2);

plot(f, angle(H_pm)*180/pi, 'b', 'LineWidth', 1.5);
hold on;
plot(f, angle(H_fs)*180/pi, 'r--', 'LineWidth', 1.5);

grid on;
xlim([0 Fs/2]);
ylim([-200 200]);

legend('Parks-McClellan', ...
    'Frequency-Sampling', 'Location', 'northeast');

title('Phase Response Comparison');
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');
```

```
=== EXPERIMENT 5: PARKS-MCCLELLAN vs FREQUENCY-SAMPLING ===

Filter Order (N): 40
Passband Edge: 1500 Hz
Stopband Edge: 2000 Hz

--- PARKS-MCCLELLAN COEFFICIENTS (First 10) ---
```

```
  Columns 1 through 7

   -0.0015    0.0072    0.0053   -0.0010   -0.0081   -0.0072    0.0037

  Columns 8 through 10

    0.0142    0.0100   -0.0090

--- FREQUENCY-SAMPLING COEFFICIENTS (First 10) ---
  Columns 1 through 7

    0.0000    0.0001    0.0002   -0.0001   -0.0008   -0.0009    0.0008

  Columns 8 through 10

    0.0035    0.0029   -0.0039
```



Published with MATLAB® R2026b